

Taking the **WEIGHT** off your **LAP**

Wireless technology is seeing a new era, thanks to the latest mobile processors. We take a peek into what makes them tick

The world of mobile computing has seldom been so exciting. Not, at least, for the last 3 years when all that the chip giants could think of was scaling down the frequency and voltage of the desktop CPUs, and labelling them as mobile processors. At CeBIT, the annual trade show for information and telecommunications technology held in Germany, Intel displayed a paradigm shift in thinking.

Intel introduced a combination of mobile technologies under the brand name Centrino (formerly Banias). Centrino includes a newly-designed microprocessor, integrated wireless capability, as well as related chipsets and software.

It can be used to make thinner and lighter notebook computers with built-in wireless capability, extended battery life and improved performance. AMD too sprang a surprise with 12 mobile processors (*See box, 'AMD is a tough competitor'*).

Given the longer batter life, frequent fliers can check their e-mail or read a newspaper online while waiting for a flight, and still have enough battery life left to watch a DVD on the return flight. Collegians, as done overseas, can register for next semester's classes, or seek a part-time job from the college library—all on a sleek and light system that won't stifle them.

How it works

Centrino broadly comprises three parts—a microprocessor called the Pentium-M, a companion chipset and a WiFi module. The Pentium-M comes out of Intel's labs in Israel and attempts to address one of the key problems with Pentium notebooks—the excessive power consumption.

The Pentium-M is designed from the ground up to fit into notebooks, i.e., the chip is built from an architecturally different point of view. It now features the Micro Ops Fusion that combines two micro-operations, enabling it to execute tasks faster, while using less power. The first micro-operation is the Advanced Instruction Prediction, aimed at reducing